Math review

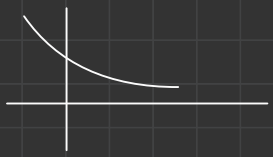
Laplace Transform

$\mathcal{L}[f(t)] = \int_0^\infty f(t)\, e^{-st}\, dt$

Most common used function.

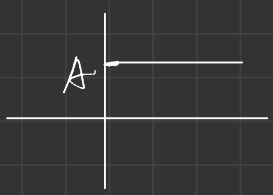
1. Exponential (指数)

$$f(t) = \begin{cases} 0 & t<0 \\ Ae^{-at} & t\geq 0 \end{cases} \Rightarrow F(s) = \frac{A}{s+a}$$



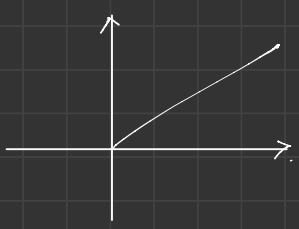
2. Step

$$f(t) = \begin{cases} 0 & t<0 \\ A & t\geq 0 \end{cases} \Rightarrow F(s) = \frac{A}{s}$$



3. Ramp

$$f(t) = \begin{cases} 0 & t<0 \\ At & t\geq 0 \end{cases} \Rightarrow F(s) = \frac{A}{s^2}$$



4 Sinusoidal.

$\mathcal{L}[A\sin\omega t] = \frac{A\omega}{s^2+\omega^2}$ $\mathcal{L}[A\cos\omega t] = \frac{As}{s^2+\omega^2}$

5. Impulse

$$\delta(t) = \begin{cases} \lim_{t_0\to 0} \frac{A}{t_0} & 0<t\leq t_0 \\ 0 & t<0,\ t>t_0 \end{cases} \qquad \mathcal{L}[\delta(t)] = A$$

Properties of.

Differential theorem. (微分定理) $\mathcal{L}[f(t)] = F(s)$

$$\mathcal{L}\left[\frac{d^n f(t)}{dt^n}\right] = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots$$

Example: $\mathcal{L}\left[\frac{d^2 f(t)}{dt^2}\right] = s^2 F(s) - s f(0) - f'(0)$

Final value theorem (终值定理)

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

Note: Final value theorem can only be used if $\lim_{t\to\infty} f(t)$ exists.

Inverse Laplace Transform.

$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} F(s) e^{-st}\, ds.$$

Partial Fraction Expansion (部分分式展开)

Case A: real and distinct poles. (单实数极点)

$$F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p_1)(s+p_2)\cdots} = \frac{k_1}{s+p_1} + \frac{k_2}{s+p_2} + \frac{k_n}{s+p_n} + \cdots$$

$$k_n = (s+p_n)F(s)\Big|_{s=-p_n}$$

Case B real, repetitive poles (重实根)

$$F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p)^r} = \frac{k_1}{s+p} + \frac{k_2}{(s+p)^2} + \cdots + \frac{k_r}{(s+p)^r}$$

系数 $k_r = (s+p)^r F(s)\Big|_{s=-p}$

$$k_{r-1} = \frac{1}{1!}\frac{d}{ds}\left[(s+p)^r F(s)\right]\Big|_{s=-p}$$

$$\cdots$$

$$k_1 = \frac{1}{(r-1)!}\frac{d^{r-1}}{ds^{r-1}}\left[(s+p)^r F(s)\right]\Big|_{s=-p}$$

Case C : Complex conjugate poles. (复数共轭极点)

$$\mathcal{L}[A e^{-at}\cos\omega t] = A\frac{s+a}{(s+a)^2+\omega^2}$$

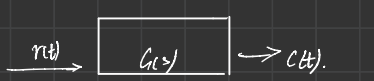
$$\mathcal{L}[A e^{-at}\sin\omega t] = A\frac{\omega}{(s+a)^2+\omega^2}$$

Example. $F(s) = \frac{s+2}{s^2+2s+5}$

$$\mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1}\left[\frac{(s+1)+1}{(s+1)^2+2^2}\right] = \mathcal{L}^{-1}\left[\frac{s+1}{(s+1)^2+2^2} + \frac{1}{2}\frac{2}{(s+1)^2+2^2}\right]$$

$$= e^{-t}\cos 2t + \frac{1}{2}e^{-t}\sin 2t$$

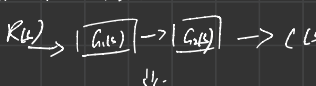
Transfer function (传递函数)

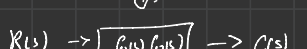


$$G(s) = \frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{Input}]}\Big/ \text{without initial conditions.}$$
(无初值.)

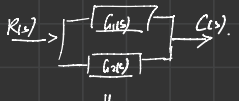
Cascode. Combinations (级联)

1) Series. 串联.





2). Parallel 并联.



$G(s) = G_1 + G_2$.

Modeling.

- Mechanical. (translational / rotational)
- Electrical. (magnetic / electrical)
- Mechatronic

General

— Common approach based energy / power
— Lumped parameter (集中参数法)
— Look for 2 power variables.

Power = f × v = τ × w = u × i

Mechancial (translational)

Power = f × v
↑ force, ↑ velocity

displacement (位移) $x(t) = \int_0^t v(t)\,dt$ $v = \frac{dx}{dt}$

momentum 冲量

$p(t) = \int_0^t f(t)\,dt$, $f = \frac{dp}{dt}$

Energy in basic elements:

Spring. $\Delta W = f(v\,dt) = f\,dx = d\mathcal{E}_{potential}$ — 势能
Mass $\Delta W = v(f\,dt) = v\,dp = d\mathcal{E}_{kinetic}$ — 动能
Damper 阻尼: $\Delta W = (vf)\,dt = d\mathcal{E}_{dissipative}$ 耗散

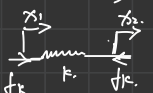
Constitutive Equation. (本构方程)

Ideal mass: rigial body (刚体)



$f = m\frac{d^2x}{dt^2}$

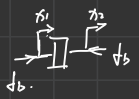
Ideal spring: no mass, no loss., linear



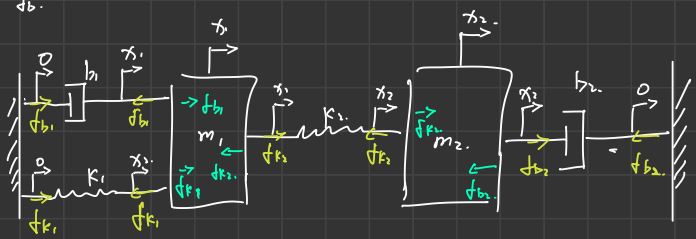
head ↑ tail

$f_k = k(x_1 - x_2)$

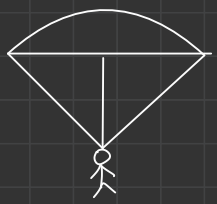
Ideal damper : no mass

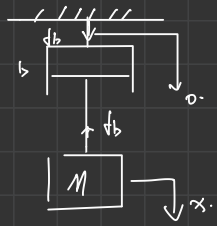


$f_b = b(\dot{x}_1 - \dot{x}_2)$


x_1
x_2
b_1
x_1
f_{b_1}
f_{b_1}
f_{b_1}
m_1
f_{k_1}
x_1
x_2
f_{k_2}
f_{k_2}
f_{k_2}
m_2
x_2
b_2
f_{b_2}
f_{b_2}
f_{k_1}
f_{k_1}
f_{k_1}
f_{k_2}
f_{b_2}
f_{b_2}
f_{k_2}
f_{b_2}
f_{k_2}
f_{k_2}
$f_{k

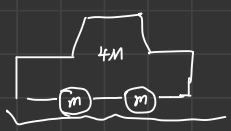
Example : Parachute (降落傘)

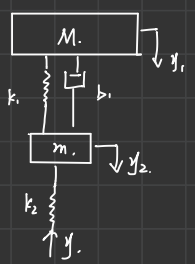




$$m\ddot{x} + b\dot{x} = 0$$

Example : $\frac{1}{4}$ Car model

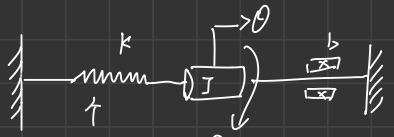




$$\frac{Y_1(s)}{Y(s)} = -\frac{k_2(b_1 s + k)}{(ms^2 + bs + k_1 + k_1)(Ms^2 + b_1 s + k_1) - (b_1 s + k_1)^2}$$

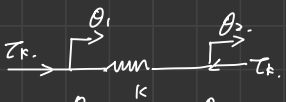
$$\frac{Y_2(s)}{Y(s)} = -\frac{k_2(Ms^2 + b_1 s + k_1)}{}$$

Mechanical (Rotational)

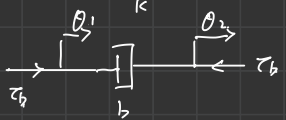


torsional spring.

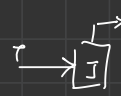
Ideal Elements



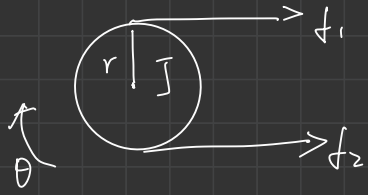
$\tau_k = k(\theta_1 - \theta_2)$



$\tau_b = b(\dot{\theta}_1 - \dot{\theta}_2)$

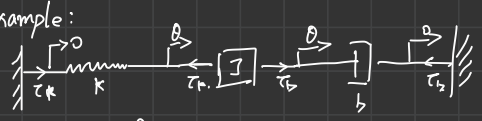


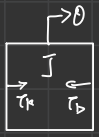
$\tau = J\ddot{\theta}$



$J\ddot{\theta} = r(f_1 - f_2)$

Example:

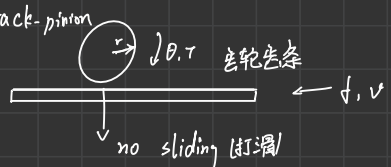




$$\left.\begin{aligned} \tau_k &= -k\theta \\ \tau_b &= b\dot{\theta} \\ \tau_k - \tau_b &= J\ddot{\theta} \end{aligned}\right\} \Rightarrow J\ddot{\theta} + b\dot{\theta} + k\theta = 0.$$

Two - port System

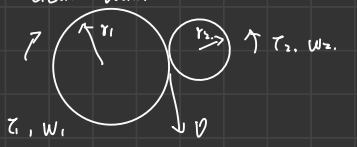
rack-pinion



$v = \omega r = \dot{\theta} r$

$w\tau = fv$

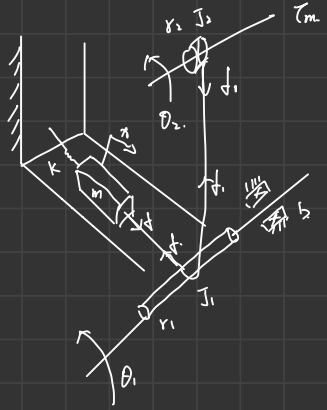
Gear-train.



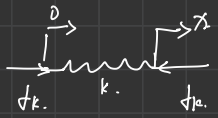
$v = w_1 r_1 = w_2 r_2.$

$w_1 \tau_1 = w_2 \tau_2.$

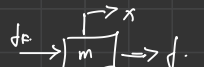
Example:



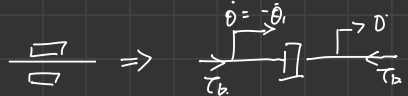
Find $\frac{X(s)}{\tau_m(s)}$



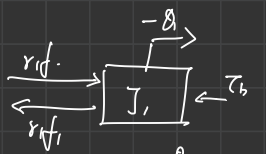
$f_k = k(0 - x) = -kx.$



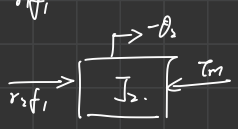
$m\ddot{x} = f + f_k.$



$\tau_b = -b\dot{\theta}_1$



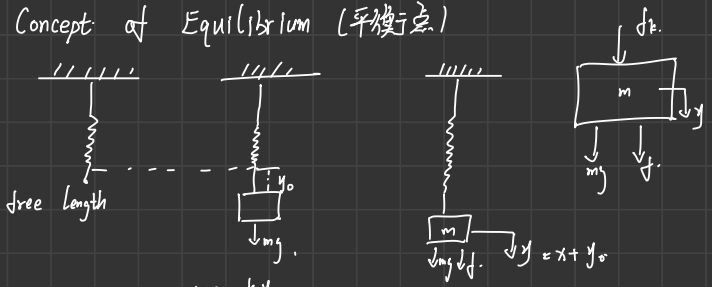
$\Sigma \tau = r_1 f - r_1 f' - \tau_b = J(-\ddot{\theta}_1)$



$\Sigma = r_2 f' - \tau_m = J_2(-\ddot{\theta}_2)$

kinetics: $x = \theta_1 r_1 = \theta_2 r_2 \Rightarrow \left(m + \frac{J_1}{r_1^2} + \frac{J_2}{r_1^2 r_2^2}\right)\ddot{x} + \frac{b}{r_1^2}\dot{x} + kx = \frac{\tau_m}{r_2}$

Concept of Equilibrium (平衡点)



$mg = k y_0$

$mg + f + f_k = m\ddot{y}$

$f_k = k(0 - y) = -ky$

$\Rightarrow m\ddot{y} + ky = f + mg$

代入 $y = x + y_0$

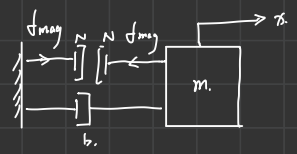
$\Rightarrow m\ddot{x} + ky_0 + kx = f + mg$

$m\ddot{x} + kx = f$

y_0 : equilibrium.

x is measured from y_0

Linearization (线性化)



$f_{mag} \approx -\frac{1}{x^2}$

$\Rightarrow m\ddot{x} + b\dot{x} + \frac{1}{x^2} = 0$

For a nonlinear system.

$y = f(x)$

If the system operates in a small region around $\bar{x}$. (for example, equilibrium) the dynamic equation can be linearized by expanding the nonlinear terms into Tylor Series (泰勒级数)

$$y = f(x) = f(\bar{x}) + \frac{df}{dx}\Big|_{x=\bar{x}} (x - \bar{x}) + \underbrace{\frac{1}{2!}\frac{d^2f}{dx^2}\Big|_{x=\bar{x}} (x - \bar{x})^2 + \cdots}_{\text{high order terms} = 0}$$

Example.

$f(x) = \frac{1}{x^2}$ around $\bar{x} = 1$

$$f(x) = f(1) + \frac{d(\frac{1}{x^2})}{dx}\Big|_{x=\bar{x}} (x-1)$$

$$= 1 + \left(-\frac{2}{x^3}\right)\Big|_{x=\bar{x}} (x-1)$$

$$= 1 + (-2)(x-1) = -2x+3$$

$$y = f(x_1, x_2)$$

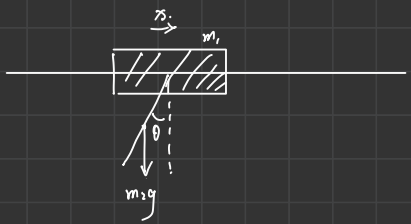
$$= f(\bar{x}_1, \bar{x}_2) + \frac{\partial f}{\partial x_1}\Big|_{\substack{x_1=\bar{x}_1 \\ x_2=\bar{x}_2}} (x_1 - \bar{x}_1) + \frac{\partial f}{\partial x_2}\Big|_{\substack{x_1=\bar{x}_1 \\ x_2=\bar{x}_2}} (x_2 - \bar{x}_2)$$

Example: $z = xy$, $5 \le x \le 7$, $10 \le y \le 12$

choose: $\bar{x} = 6$, $\bar{y} = 11$

$$z = \bar{x}\bar{y} + \frac{\partial(xy)}{\partial x}\Big|_{\substack{x=6 \\ y=11}} (x-6) + \frac{\partial(xy)}{\partial y}\Big|_{\substack{x=6 \\ y=11}} (y-11) = 11x + 6y - 66$$

Dynamic of Multi-Body Systems



Equation of motion (Dynamic equation)

Approach.

① Newton-Eulev Method 1)物理过程更清晰 2)考虑内力

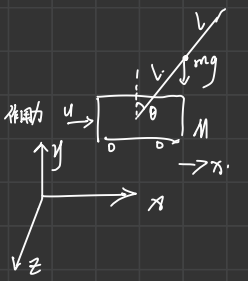
② Lagrange Formulation. 1)无需考虑内力 2)中间过程不清晰

1) Newton - Euler.

$\Sigma F = ma = m\ddot{x}$

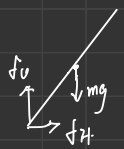
$\Sigma \tau = J\ddot{\theta}$

Example: Inverted Pendulum (倒立擺)



Free body diagram:

Pendulum:



center of gravity

$$\begin{cases} x_G = x + l\sin\theta \\ y_G = l\cos\theta \end{cases}$$

(1) Σf in horizontal (水平)

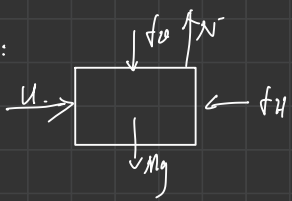
$$f_H = m\ddot{x}_G = m\frac{d}{dt}\left(\frac{dx_G}{dt}\right)$$
$$= m\frac{d}{dt}\left(\dot{x} + l\dot{\theta}\cos\theta\right)$$
$$= m\left[\ddot{x} + l\ddot{\theta}\cos\theta - l\sin\theta\,\dot{\theta}^2\right] \quad ①$$

(2) Σf in vertical

$$f_v - mg = m\ddot{y}_G = -ml\left[\cos\theta\cdot\dot{\theta}^2 + \sin\theta\,\ddot{\theta}\right] \quad ②$$

(3) $\Sigma\tau = f_v l\sin\theta - f_H l\cos\theta = J\ddot{\theta}$ ③

Car:



Σf in. horizontal:

$$u - f_H = M\ddot{x}. \quad ④$$

Σf in. vertical:

$$f_v + Mg - N = 0 \quad ⑤$$

(1) & (4) $\Rightarrow$ $(M+m)\ddot{x} + ml\cos\theta\cdot\ddot{\theta} - ml\sin\theta\,\dot{\theta}^2 = u.$

(1)(2)(3) $\Rightarrow$ $(J + ml^2)\ddot{\theta} + ml\cos\theta\,\ddot{x} - mgl\sin\theta = 0$

2> Lagrange Formulation:

variable in generalized coordinate (广义坐标)

$\vec{q} = [\vec{r}, \vec{\alpha}]^T$ eg. $\vec{q} = [\vec{r}, \vec{\theta}]$, $\vec{q} = [q_1, \cdots q_j \cdots]$

$\vec{r}$: displacement; $\vec{\alpha}$: angular displacement.

Lagrange Equation

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} = Q_j \quad j = 1, 2, \cdots$$

T: kinetic energy (动能) — translation; rotation.

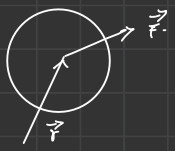
U: potential energy (势能) — gravity; spring.

Q_j: generalize force 广义力

virtual displacement 虚位移

$$\delta \vec{r} = \sum_{j=1}^{N} \frac{\partial \vec{r}}{\partial q_j} \delta q_j$$

The virtual displacement can be also obtained be replacing $\dot{q}_i$ with δq_i in the velocity expression



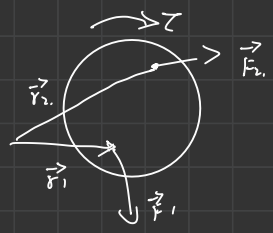
$$\delta \vec{\alpha} = \sum_{j=1}^{N} \vec{w}_j \delta q_j$$

virtual works 虚功

$$\delta W = \sum \vec{F}_n \cdot \delta \vec{r}_n + \sum \vec{\tau}_n \cdot \delta \vec{\alpha}_n$$

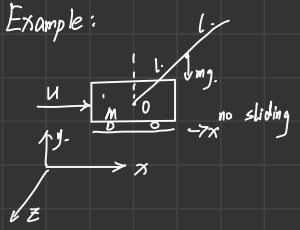
$$= \sum Q_j \cdot q_j$$

广义力



(w) → τ (外力矩)

Example:



$\vec{q} = [x, \theta]^T = [q_1, q_2]^T$

动能 $T = \frac{1}{2}MV_0^2 + \frac{1}{2}mV_A^2 + \frac{1}{2}J\dot{\theta}^2$

势能 $V = mgl\cos\theta$

$\vec{v}_0 = \dot{x}\vec{i}$

$\vec{v}_A = \vec{v}_0 + \vec{\omega}_A \times \vec{r}_{OA}$

$= \dot{x}\vec{i} + \dot{\theta}(-\vec{k}) \times [l\sin\theta\,\vec{i} + l\cos\theta\,\vec{j}]$

$= [\dot{x} + (\dot{\theta}\cos\theta)]\vec{i} - l\dot{\theta}\sin\theta\,\vec{j}$

$T = \frac{1}{2}M\vec{V}_0^T\vec{V}_0 + \frac{1}{2}m\vec{V}_A^T V_A + \frac{1}{2}J\dot{\theta}^2$

$= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}J\dot{\theta}^2$

虚位移:

(外力作用点) $\delta\vec{r}_0 = \frac{\partial\vec{r}_0}{\partial q_1}\delta q_1 + \frac{\partial\vec{r}_0}{\partial q_2}\delta q_2$

$= \frac{\partial\vec{r}_0}{\partial x}\delta x + 0\,\delta\theta$

$= \delta x \cdot \vec{i}$

或 $\vec{v}_0 = \dot{x}\vec{i}$ 用 $\delta\dot{x}$ 来代替 则 $\delta\vec{r}_0 = \delta x\,\vec{i}$

虚功 $\delta W = \vec{U}\cdot\delta\vec{r}_0 = (U\vec{i})\cdot(\delta x\vec{i}) = U\delta x = \sum_{j=1}^{2} Q_j q_j = Q_1 q_1 + Q_2 q_2$

($\dot{x}$ 对应)

$\Rightarrow \begin{cases} Q_1 = U \\ Q_2 = 0 \end{cases}$

$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = Q_j$

$j=1,\ q_1 = x$ $\Rightarrow \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} = Q_1 = U$

$\Rightarrow (M+m)\ddot{x} + ml\cos\theta\cdot\ddot{\theta} - ml\sin\theta\cdot\dot{\theta}^2 = U$

$j=2,\ q_2 = \theta$ $\Rightarrow \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}}\right) - \frac{\partial T}{\partial \theta} + \frac{\partial V}{\partial \theta} = Q_2 = 0$

$\Rightarrow (J + ml^2)\ddot{\theta} + ml\cos\theta\,\ddot{x} - mgl\sin\theta = 0$

线性化 $(M+m)\ddot{x} + ml\ddot{\theta} = U$

$(J + ml^2)\ddot{\theta} + ml\ddot{x} - mgl\theta = 0$

Free response . Forced response.

Transient response (瞬态响应).

steady-state response (稳态响应)



Dynamic equation :

$m\ddot{x} + kx = f$.

Take L.T.

$\Rightarrow m[s^2X(s) - sx(0) - \dot{x}(0)] + kX(s) = F(s)$

$\Rightarrow X(s) = \underbrace{\frac{\dot{x}(0)}{s^2+\frac{k}{m}} + \frac{sx(0)}{s^2+\frac{k}{m}}}_{\text{1. initial condition.}} + \underbrace{\frac{F(s)}{ms^2+k}}_{\text{2. input.}}$

1> Free response , $f = 0$, $x(0) \neq 0$, $\dot{x}(0) \neq 0$

$X(s) = \frac{\dot{x}(0)}{s^2+\frac{k}{m}} + \frac{sx(0)}{s^2+\frac{k}{m}}$

Inv L.T. $\Rightarrow$

$x(t) = \sqrt{\frac{m}{k}}\,\dot{x}(0)\sin\sqrt{\frac{k}{m}}\,t + x(0)\cos\sqrt{\frac{k}{m}}\,t$

$= A\sin(\sqrt{\frac{k}{m}}\,t + \phi)$

$w_n = \sqrt{\frac{k}{m}}$, natural frequency

2> Forced response. , $f \neq 0$, $x(0) = \dot{x}(0) = 0$ $\quad f = P\sin wt$

$X(s) = \frac{P}{m}\frac{w^2}{s^2+w^2} \cdot \frac{1}{s^2+\frac{k}{m}}$

$= \frac{-Pw\sqrt{\frac{m}{k}}}{k-mw^2}\frac{\sqrt{\frac{k}{m}}}{s^2+\frac{k}{m}} + \frac{P}{k-mw^2}\frac{w}{s^2+w^2}$

$x(t) = \underbrace{\frac{-Pw\sqrt{\frac{m}{k}}}{k-mw^2}\sin\sqrt{\frac{k}{m}}\,t}_{\downarrow w_n} + \frac{P}{k-mw^2}\sin wt$

这一项会因阻尼随时间衰减

The first term decays with time. when there is damping 阻尼, this term is called transient response.

$x(t)\big|_{t>t_c} = \frac{P}{k-mw^2}\sin wt$ $\quad t_c$: criticed time.

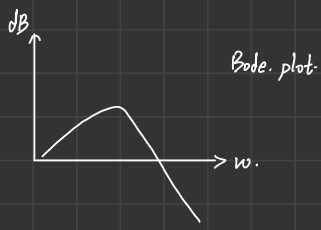
steady-state response.

when $K - mw^2 = 0$, $w = w_n = \sqrt{\frac{K}{m}}$

$x(t) \to \infty$ resonance (共振)

$X(t) = \frac{1}{K - mw^2} \cdot \sin wt$

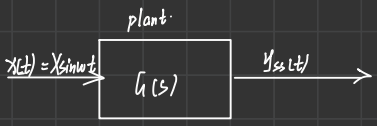
$\frac{1}{K-mw^2}$: magnitude. $\sin wt$: input



Bode. plot.

Frequency response.

steady - state response of a system to a. sinusoidal input: 正弦



Goal: System Identification. Find G(s)

1. Modeling: knowledge. about the system is required

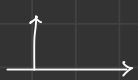


2. Experimental:



$\Rightarrow G(s) = \frac{Y(s)}{X(s)}$

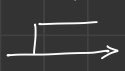
① Impulse response.



$X(s) = 1$

$\Rightarrow G(s) = Y(s)$

② Step. response.

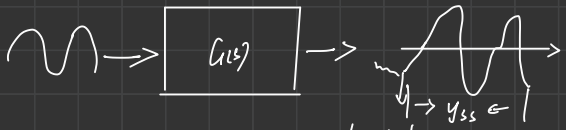


$X(s) = \frac{1}{s}$

$G(s) = s\,Y(s)$

① and ② may not be implementable in practice.

③ Frequency response.



transient response

$x(t) = X\sin wt$ → $G(s)$ → $y(t),\ y_{ss}(t)$.

L.T ↓

$$X(s) = \frac{Xw}{s^2+w^2} = \frac{Xw}{(s+jw)(s-jw)}$$

Suppose. $G(s) = \frac{P(s)}{Q(s)} = \frac{P(s)}{(s+s_1)(s+s_2)+\cdots}$

$$\Rightarrow Y(s) = X(s)G(s) = \frac{Xw}{(s+jw)(s-jw)} \cdot \frac{P(s)}{(s+s_1)(s+s_2)\cdots}$$
$$= \frac{a}{s+jw} + \frac{\bar{a}}{s-jw} + \frac{b_1}{s+s_1} + \frac{b_2}{s+s_2} + \cdots$$

Inverse L.T. $y(t) = ae^{-jwt} + \bar{a}e^{jwt} + b_1e^{-s_1t} + b_2e^{-s_2t} + \cdots$

$t \to \infty$ $\quad y_{ss}(t) = ae^{-jwt} + \bar{a}e^{jwt}$

To find a and $\bar{a}$.

$$Y(s)(s+jw) = G(s)\frac{Xw}{(s+jw)(s-jw)}(s+jw)$$
$$= a + \left(\frac{\bar{a}}{s-jw} + \frac{b_1}{s+s_1} + \cdots\right)(s+jw)$$

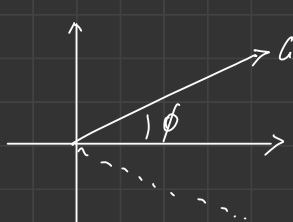
$s = -jw$

↓

$$G(-jw)\frac{Xw}{-2jw} = a = -G(-jw)\frac{X}{2j}$$

Similarly $\bar{a} = G(jw)\frac{X}{2j}$

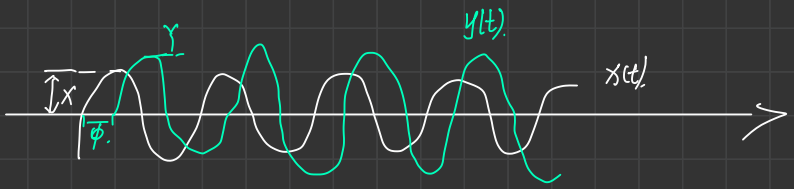
$$\Rightarrow y_{ss}(t) = X\frac{G(-jw)e^{-jwt} - G(jw)e^{jwt}}{-2j}$$



$G(jw) = |G(jw)|e^{j\phi}$

$G(-jw) = |G(jw)|e^{-j\phi}$

$$\Rightarrow y_{ss}(t) = X|G(jw)|\frac{e^{j(wt+\phi)} - e^{-j(wt+\phi)}}{2j} = X|G(jw)|\sin(wt+\phi) = Y\sin(wt+\phi)$$



$$|G(jw)| = \frac{Y}{X} \qquad \angle G(jw) = \phi$$

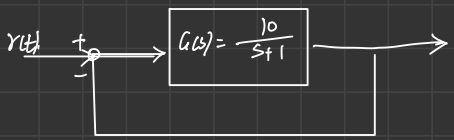
Example: Consider a unity-feedback system with open-loop transfer function. $G(s) = \frac{10}{s+1}$

Find the steady-state output for the following input

(a) $r(t) = \sin(t+30°)$

(b) $r(t) = 2\cos(2t-45°)$

(c) $r(t) = \sin(t+30°) - 2\cos(2t-45°)$



Solution: closed-loop transfer function.

$$G(s) = \frac{C(s)}{R(s)} = \frac{\frac{10}{s+1}}{1+\frac{10}{s+1}} = \frac{10}{s+11}$$

$$s = jw \Rightarrow G(jw) = \frac{10}{jw+11} \Rightarrow |G(jw)| = \frac{10}{\sqrt{121+w^2}}$$

$$\angle G(jw) = -\tan^{-1}\frac{w}{11}$$

(a) $r(t) = \sin(t+30°)$, $w=1$

$\Rightarrow |G(jw)| = \frac{10}{\sqrt{122}} = 0.0905$, $\angle G(jw) = -5.19°$

$$C_{ss}(t) = 1 \times 0.0905 \times \sin(t+30°-5.19°) = 0.0905\sin(t+24.8°)$$

(b) $r(t) = 2\cos(2t - 45°)$, $w = 2$.

$|G(jw)| = \frac{10}{\sqrt{125}} = 0.89$ $\quad \angle G(jw) = -10.3°$

$C_{ss}(t) = 2 \times 0.89 \cos(2t - 45° - 10.3°) = 1.78 \cos(2t - 55.3°)$.

(c) $r(t) = \sin(t + 30°) - 2\cos(2t - 45°)$.

$C_{ss}(t) = 0.0905 \sin(t + 24.8°) - 1.78 \cos(2t - 55.3°)$.

Electrical System

Power variables:

Power = e × i
(e: voltage, i: current)

Other variables:

electrical charge (电量, coulomb)

$q(t) = \int_0^t i(t)\,dt$

magnetic flux linkage (磁通链, wb)

$\lambda(t) = \int_0^t e(t)\,dt$, $e(t) = \frac{d\lambda(t)}{dt}$

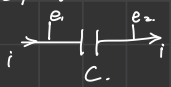
$\Delta W = i\,e(t)\,dt = i\,d\lambda = d\mathcal{E}_{magnetic}$ → spring-like.

$\Delta W = e(t)\,i(t)\,dt = e\,dq = d\mathcal{E}_{electrical}$ → mass-like.

$\Delta W = i\,e\,dt = d\mathcal{E}_{resistive}$ → damper-like.

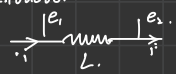
Ideal Elements

Capacitor:



$i(t) = C\frac{d}{dt}(e_1 - e_2)$

Inductor



$i(t) = \frac{1}{L}\int_0^t (e_1 - e_2)\,dt$

or $e_1 - e_2 = L\frac{di(t)}{dt}$

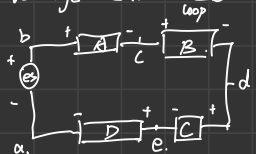
Resistor:



$e_1 - e_2 = iR$

Physical Laws

1) Voltage Law: $\sum_{loop} e = 0$



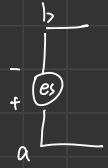
define: volt ↑ ⇒ "+"

volt ↓ ⇒ "−"

For voltage source:



$e_{ab} = +e_s$

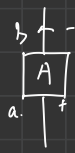


$e_{ab} = -e_s$.

For ideal elements



$e_{ab} = +e_A = +iR = +L\frac{di}{dt} = \frac{1}{C}\int i\,dt$



$e_{ab} = -e_A = -iR = -L\frac{di}{dt} = -\frac{1}{C}\int i\,dt$.

2) Current Law $\sum_{\text{node}} i = 0$

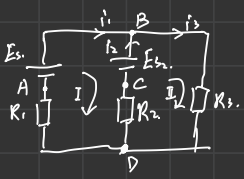


$i_{enter} \Rightarrow$ "+"

$i_{leave} \Rightarrow$ "−"

$\sum i = i_a + i_b + i_c - i_d = 0$

Example:



Current Law:

at node B : $\Sigma i = i_1 + i_2 - i_3 = 0$

Voltage Law:

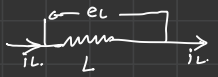
Loop I: $\Sigma e = e_{AB} + e_{BC} + e_{CD} + e_{DA}$

$= E_{s1} - E_{s2} + i_2R_2 - i_1R_1 = 0$

Loop II $\Sigma e = e_{CB} + e_{BD} + e_{DC}$

$= E_{s2} - i_3R_3 - i_2R_2 = 0$

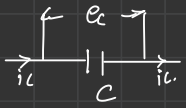
Transfer function. (zero initial condition)



$e_{L(t)} = L \frac{d\, i_{L(t)}}{dt}$

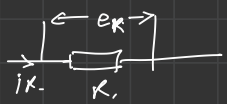
$E_{L(s)} = L s I_{L(s)}$

$\frac{E_{L(s)}}{I_{L(t)}} = Ls = Z_{(s)}$. complex impedance (阻抗)



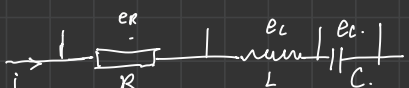
$i_{C(t)} = C \frac{de_{C(t)}}{dt}$

$I_{o(s)} = Cs\, E_{C(s)}$. $\Rightarrow Z_{(s)} = \frac{E_{L(s)}}{I_{C(s)}} = \frac{1}{Cs}$



$Z_{(s)} = \frac{E_{R(s)}}{I_{R(s)}} = R$.

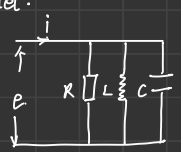
series



$Z_{(s)} = \frac{E_R + E_L + E_C}{I} = Z_R + Z_L + Z_C$.

$= R + LS + \frac{1}{Cs}$

Parallel:

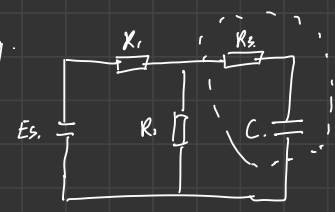


$Z_{(s)} = \frac{E}{I} = E$

$I = \frac{E}{Z_R} + \frac{E}{Z_L} + \frac{E}{Z_C}$

$\Rightarrow Z_{(s)} = \frac{E}{I} = \frac{1}{\frac{1}{Z_R} + \frac{1}{Z_L} + \frac{1}{Z_C}}$

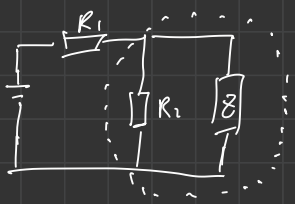
Eg.



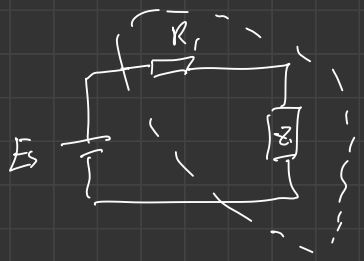
$Z = Z_{R3} + Z_C$

$= R_3 + C_s$.

$\Rightarrow$

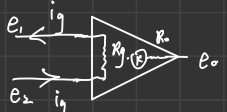


$Z_T = \frac{1}{\frac{1}{R_1} + \frac{1}{Z}}$



$$Z_2 = Z_{R_1} + Z_1 = \frac{E}{I}$$

Electrical Operational Amplifier (运放)



Ideal. Op-amp.

$R_g \to \infty$, $R_0 \to 0$

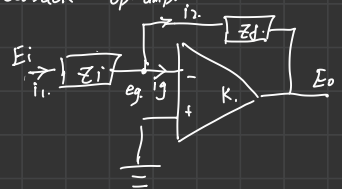
$\Rightarrow i_g = 0$ ①

$e_0 = k(e_2 - e_1)$

$k \approx 10^4 \sim 10^8$

$\Rightarrow e_1 = e_2$. ②

Feedback. op-amp.



Z_i : input impedance.

Z_f : feedback impedance.

$\frac{E_0}{E_i} = ?$

$i_g = 0 \Rightarrow i_1 = i_2$.

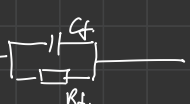
$e_g = 0$.

$I_1 = \frac{E_i - \overset{0}{E_g}}{Z_i} = I_2 = \frac{\overset{0}{E_g} - E_0}{Z_f}$

$\Rightarrow \frac{E_0}{E_i} = -\frac{Z_f}{Z_i}$

Example :

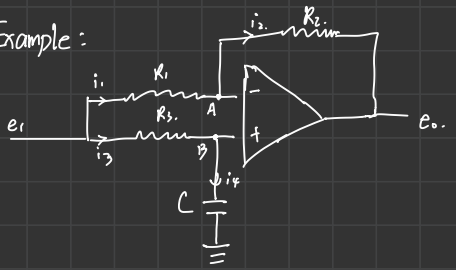
① $\begin{cases} Z_i = R_i \\ Z_f = R_f \end{cases} \Rightarrow \frac{E_0}{E_i} = -\frac{R_f}{R_i}$

② $Z_f =$ 

$$Z_f = \frac{1}{\frac{1}{Z_{Cf}} + \frac{1}{Z_{Rf}}} = \frac{1}{\frac{1}{R_f} + C_f s} = \frac{R_f}{1 + R_f C_f s}$$

$Z_i = R_i \Rightarrow \frac{E_0}{E_i} = -\frac{Z_f}{Z_i} = \frac{R_f}{R_i(1 + R_f C_f s)}$

Example:



$\frac{E_0}{E_i} = ?$

$E_A = E_B$, $I_1 = I_2$, $I_3 = I_4$

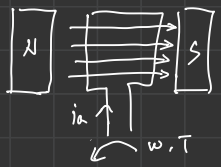
$I_1 = \frac{E_i - E_A}{R_1} = I_2 = \frac{E_A - E_0}{R_2}$

$I_3 = \frac{E_i - E_B}{R_3} = I_4 = \frac{E_B - 0}{\frac{1}{Cs}}$

$\Downarrow$

$\frac{E_0}{E_i} = \frac{R_1 + R_2}{R_1(1 + R_2 Cs)} - \frac{R_2}{R_1}$

Motor / Generator



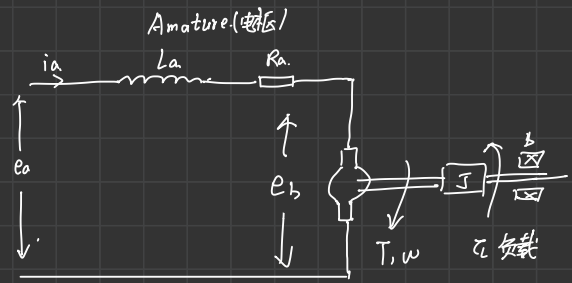
Motor

$T = K_a i_a$

$e_b = K_b \omega$

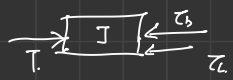
反电动势 back-emf.

ideal : $K_a \approx K_b$



Electrical system : Motor $T = K_a i_a$, $e_b = K_b \omega$: Mechanical system.

$$e_a - L_a \frac{di_a}{dt} - R_a i_a - e_b = 0$$



$$J\dot{\omega} = T - \tau_b - \tau_L$$
$$= T - b\omega - \tau_L$$

$$W(s) = G_1(s) E_a(s) + G_2(s) T_L(s)$$

$$G_1(s) = \frac{W(s)}{E_a(s)}\Big|_{\tau_L = 0}, \quad G_2(s) = \frac{W(s)}{E_a(s)}\Big|_{E_a = 0}$$

1) when $\tau_L = 0$, Find $\frac{W(s)}{E_a(s)}$

Mechanical :

$$J\dot{\omega} = T - b\omega = K_a i_a - b\omega$$
$$\Rightarrow J\dot{\omega} + b\omega = K_a i_a$$

L.T. $\Rightarrow (Js + b) W(s) = K_a I_a(s)$

$$\Rightarrow \frac{W(s)}{I_a(s)} = \frac{K_a}{Js + b} \quad ①$$

Electrical :

$$e_a = e_b + L_a \frac{di_a}{dt} + R_a i_a$$
$$= K_b \omega + \cdots$$

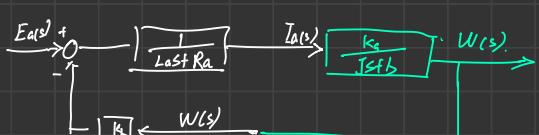
L.T. $\Rightarrow$

$$E_a(s) = K_b W(s) + L_a s I_a(s) + R_a I_a(s)$$
$$= K_b W(s) + (L_a s + R_a) I_a(s)$$
$$\Rightarrow I_a(s) = \frac{E_a(s) - K_b W(s)}{L_a s + R_a} \quad ②$$

We can solve for $\frac{W(s)}{E_a(s)}$ directly with ① ②

Alternatively : block - diagram.

② $\Rightarrow$



$$\frac{W(s)}{E_a(s)} = \frac{\frac{1}{L_a s + R_a} \cdot \frac{K_a}{Js + b}}{1 + \frac{1}{L_a s + R_a} \cdot \frac{K_a}{Js + b} \cdot K_b}$$

$$= \frac{K_a}{L_a J s^2 + (R_a J + L_a b) s + R_a b + K_a K_b}$$

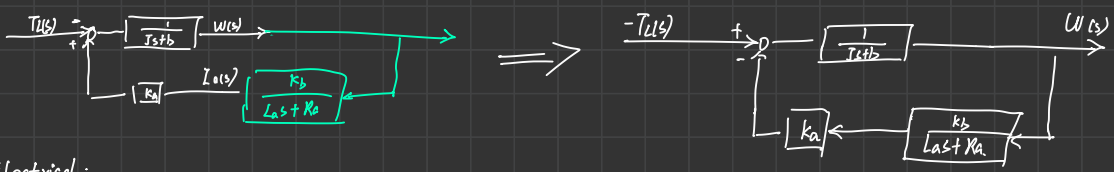
27 $e_a = 0$ Find $\frac{W(s)}{T_L(s)}$

Mechanical:

$$J\dot{w} = T - bw - T_L$$
$$= k_a i_a - bw - T_L$$

L.T. $\Rightarrow$

$$Js\,W = k_a I_a(s) - b\,W(s) - T_L(s)$$
$$W(s) = \frac{1}{Js+b}\left[k_a I_a(s) - T_L(s)\right].$$



Electrical:

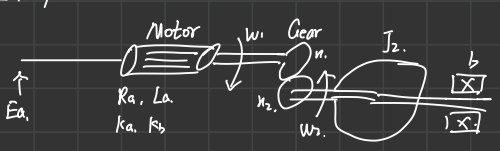
$$e_b + L_a\frac{di_a}{dt} + R_a i_a = 0$$
$$k_b w + L_a\frac{di_a}{dt} + R_a i_a = 0.$$

L.T. $\Rightarrow$

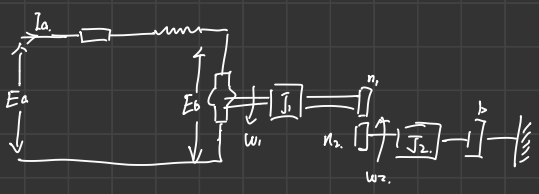
$$k_b W(s) + L_a s I_a(s) + R_a I_a(s) = 0$$
$$\Rightarrow I_a(s) = \frac{-k_b}{L_a s + R_a} W(s)$$

$$\frac{W(s)}{T_L(s)} = \frac{L_a s + R_a}{L_a J s^2 + (R_a J + L_a b)s + R_a b + k_a k_b}$$

Example:



Find $\frac{W_2(s)}{E_a(s)}$ =



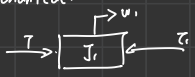
Electrical :

$$e_a - i_a R_a - L_a \frac{di_a}{dt} - e_b = 0$$

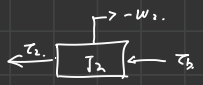
Motor : $T = k_a i_a$

$$e_b = k_b \omega_1$$

Mechanical:



$$T - \tau_1 = J_1 \dot{\omega}_1$$



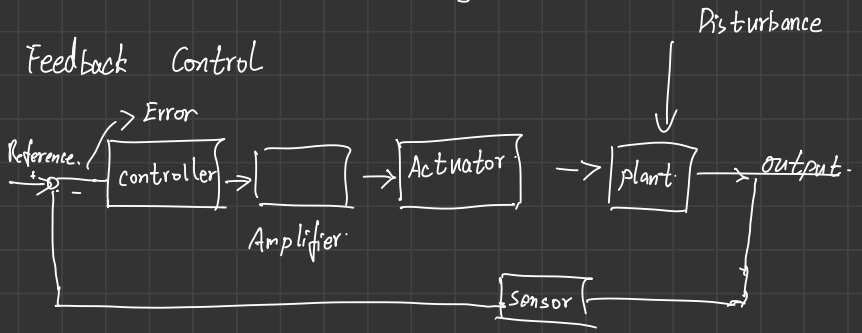
$$J_2 \dot{\omega}_2 = \tau_2 - b\omega_2$$

$$\frac{\tau_1}{\tau_2} = \frac{n_1}{n_2} = \frac{\omega_2}{\omega_1} = A$$

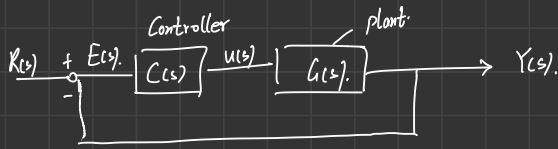
$$\Rightarrow \frac{W_2(s)}{E_a(s)} = \frac{k_a A}{(J_1 + J_2 A^2) L_a s^2 + (J_1 R_a + J_2 A^2 R_a + L_a b A^2) s + R_a b A^2 + k_a k_b A}$$

PID Control and PID Tuning

Feedback Control



Goal = drive error to zero



PID : $u(t) = k_p e(t) + k_i \int e(t)dt + k_d \frac{de(t)}{dt}$

↑ 比例项 Proportional. ↑ 积分 Integral. ↑ 微分 Derivative

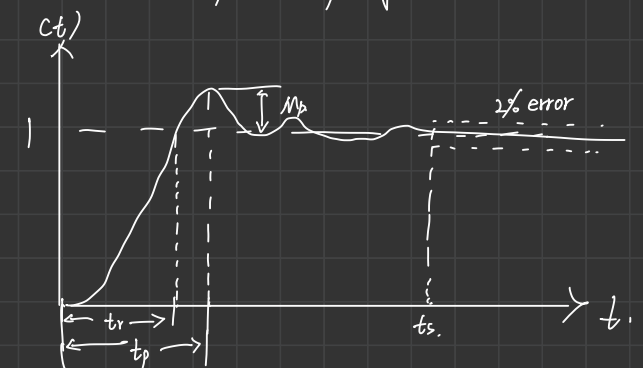
$$C(s) = \frac{U(s)}{E(s)} = k_p + \frac{k_i}{s} + k_d s$$

$$= k_p \left(1 + \frac{1}{T_i s} + T_d s\right)$$

其中 $T_i = \frac{k_p}{k_i}$, integral time

$T_d = \frac{k_d}{k_p}$, derivative time.

Transisent response specifications:



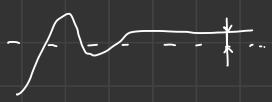
t_r : rise time. , 0 -> 100%.

t_p : peak time , 0 -> first peak.

M_p : maximum overshoot (超调)

t_s : settling time. 0 -> steady-state.

steady-state error



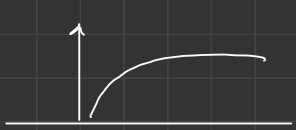
Response.	t_r / t_p	M_p.	t_s.	s-s error
k_p	↓.	↑	Small change.	↓
k_i	↓	↑	↑	Elimiate.
k_d.	Small change.	↓	↓.	No change.

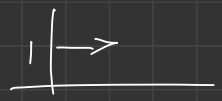
PID Tuning

Ziegler - Nichols Rules:

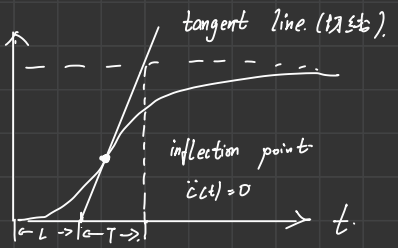
provide a starting point of the PID parameters for fine tuning.

① First Method : 1st order or over-damped (过阻尼)





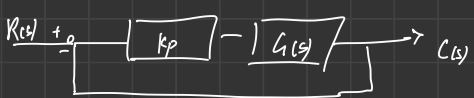




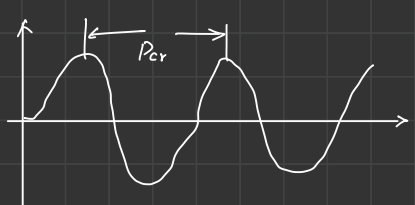
Types of Controller	k_p	T_i	T_d
P.	T/L.	∞	0
PI	$0.9T/L$.	$\frac{L}{0.3}$	0.
PID	$1.2T/L$.	$2L$	$0.5L$.

② Second Method: (2nd order and higher, under-damped). 欠阻尼

a) set: $T_i=\infty$, $T_d=0$, use only k_p



b) increase k_p, until $c(t)$ exhibits sustained oscillations (临界稳定, 等幅振荡)



$K_{cr}=k_p$

Types of Controller	k_p	T_i	T_d
P	$0.5k_{cr}$	∞	0.
PI.	$0.45k_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6k_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$.

state space. 状态空间

classical control:

— Single. input, single output.

— Scalar ODE

— Control, PID.

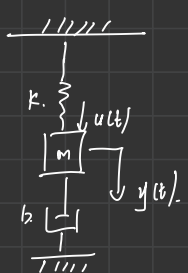
Modern control:

— Multi input, multi output. (Multi-DOF, system states: displacement, speed).

— Group of ODEs.

— Controller ?

Example:



Dynamic Equation.

$m\ddot{y} + b\dot{y} + ky = u$

state variable. 状态变量

$$\vec{x} = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} y(t) \\ \dot{y}(t) \end{bmatrix}$$

$$\Rightarrow \dot{x}_1 = \dot{y} = x_2$$

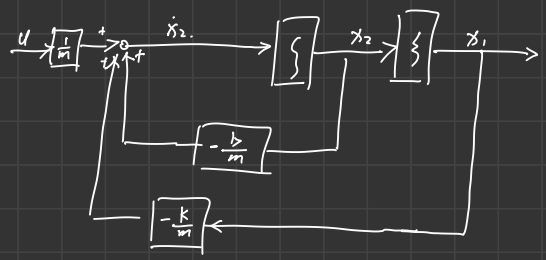
$$\dot{x}_2 = \ddot{y} = -\frac{k}{m}y - \frac{b}{m}\dot{y} + \frac{u}{m}$$

$$= -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{u}{m}.$$

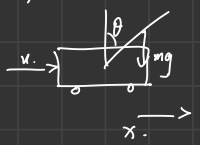
$$\Rightarrow$$

$$\dot{\vec{x}} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u.$$

$\dot{\vec{x}} = A\vec{x} + Bu$, 状态方程.



Example.



$$(M+m)\ddot{x} + ml\ddot{\theta} = u$$
$$(J+ml^2)\ddot{\theta} + ml\ddot{x} - mgl\theta = 0$$

$$\Rightarrow \begin{cases} a\ddot{x} + b\ddot{\theta} = u \\ c\ddot{\theta} + b\ddot{x} - d\theta = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \ddot{\theta} = \frac{ad}{ac-b^2}\theta - \frac{b}{ac-b^2}u = \bar{a}\theta - \bar{b}u \\ \ddot{x} = \frac{-abd}{a^2c-ab^2}\theta + \frac{ac}{a^2c-ab^2}u = -\bar{c}\theta + \bar{d}u \end{cases}$$

Let state variable:

$$\vec{x} = \begin{bmatrix} \theta \\ \dot{\theta} \\ x \\ \dot{x} \end{bmatrix} \qquad \dot{\vec{x}} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{x} \\ \ddot{x} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ \bar{a} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\bar{c} & 0 & 0 & 0 \end{bmatrix}}_{A} \vec{x} + \underbrace{\begin{bmatrix} 0 \\ -\bar{b} \\ 0 \\ \bar{d} \end{bmatrix}}_{B} u$$

Correlation between transfer function and state space.

transfer function: $\frac{Y(s)}{U(s)} = G(s)$

state space $\begin{cases} \dot{\vec{x}} = A\vec{x} + Bu \\ \vec{y} = C\vec{x} + Du \end{cases}$

take Laplace transform

$$\begin{cases} sX(s) = AX(s) + BU(s) \\ Y(s) = CX(s) + DU(s) \end{cases}$$

$\Rightarrow X(s) = (sI-A)^{-1}BU(s)$, I Identity matrix 单位矩阵

$$Y(s) = [C(sI-A)^{-1}B + D]U(s)$$

$$\Rightarrow G(s) = \frac{Y(s)}{U(s)} =$$

$$X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad X^{-1} = \frac{1}{|X|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad |X| = ad - bc.$$

$$X = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad |X| = a_1(b_2c_3 - c_2b_3) - a_2(b_1c_3 - c_1b_3) + a_3(b_1c_2 - c_1b_2)$$

$$X^{-1} = \frac{1}{|X|}\begin{bmatrix} b_2c_3 - c_2b_3 & c_1b_3 - b_1c_3 & b_1c_2 - c_1b_2 \\ c_2a_3 - a_2c_3 & a_1c_3 - c_1a_3 & a_2c_1 - a_1c_2 \\ a_2b_3 - b_2a_3 & b_1a_3 - a_1b_3 & a_1b_2 - a_2b_1 \end{bmatrix}$$

$$G(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{|sI-A|}$$

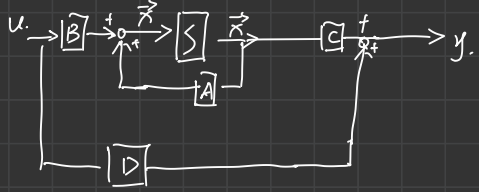
① $|sI-A|$ is characteristic polynomial of $G(s)$.
特征 多项式

② the eigenvalues of $A \Leftrightarrow$ poles of $G(s)$
特征值 极点

Controllability (可控性)

Observability (可观性).

state space $\begin{cases} \dot{\vec{x}} = A\vec{x} + Bu. \\ y = C\vec{x} + Du. \end{cases}$



$\vec{x}$: state variable. (n - vector).

$\vec{u}$: input. (r - vector)

$\vec{y}$: output (m - vector).

A $n \times n$.

B $n \times 1$

C $m \times n$.

D $m \times r$

input $\xrightarrow{u}$ [$\vec{x}(t_0) \to \vec{x}(t)$] $\xrightarrow{y}$ output

Controllability:
A system is controllable at t_0 if it is possible to find $u(t)$ to transfer the system state from $\vec{x}(t_0)$ to any other $\vec{x}(t)$ in a finite time.

find $u(t) \Rightarrow x(t_0) \to \vec{x}(t)$.
$\to \vec{x}(t)$
$\to \vec{x}(t)$ $\quad t < \infty$.

Observability:
A system is observable at t_0 if it is possible to determine $\vec{x}(t)$ from $y(t)$ in a

finite time. $y(t) \to x(t)$ $t < t_0$.

Complete state controllable.

$\text{rank}\,[B \vdots AB \vdots \cdots \vdots A^{n-1}B] = n$.

秩.

求行列式 $|\cdot| = 0$, 不满秩.

$|\cdot| \neq 0$, 满秩.

Output Controllable.

$\text{rank}\,[CB \vdots CAB \vdots CA^2B \vdots \cdots \vdots CA^{n-1}B \vdots D] = m$

Observable.

$$\text{rank}\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} = n.$$

Example: $\dot{\vec{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}}_{A} \vec{x} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} u$. $n=2$, $m=1$

$y = \underbrace{[1 \quad 0]}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $D = 0$

$\text{rank}\,[B \vdots AB] = \text{rank}\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} = 2$. $\Rightarrow$ complete state controllable.

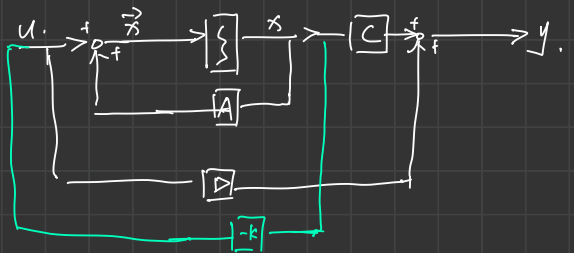
$$\text{rank}\begin{bmatrix} C \\ CA \end{bmatrix} \overset{\text{rank}}{=} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = 2. \Rightarrow \text{observable}$$

Pole Placement Method 極点配置法

— Classical control : specify dominant poles. 主.

— state feedback: specify all poles.

$$\begin{cases} \dot{\vec{x}} = A\vec{x} + Bu. \\ y = C\vec{x} + Du. \end{cases}$$



$u = -k\vec{x}$. $\quad k = [k_1, k_2, \cdots k_n]$

$$\dot{\vec{x}} = A\vec{x} + Bu = A\vec{x} - Bk\vec{x} = (A-Bk)\vec{x} = A^*\vec{x}.$$

eigenvalues of $A^* = A - Bk$ ⟺ poles

$$|sI - A^*| = |sI - A + Bk| = (s-\mu_1)(s-\mu_2)\cdots(s-\mu_n).$$

$\mu_1 \cdots \mu_n$: desired poles

Procedure of Pole Placement Method with state feedback.

step 1 : check the controllability

rank $[B : AB : \cdots : A^{n-1}B]$ if or not n

step 2 : derive the characteristic equation.

$$(s-\mu_1)(s-\mu_2)\cdots(s-\mu_n) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_{n-1}s + \alpha_n.$$

step 3 : let $|sI - A + Bk| =$ ↑ solution of $k = [k_1, \cdots k_n]$.

Example.

$\dot{\vec{x}} = A\vec{x} + Bu$, $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

place the desired poles at:

$s_1 = -2+4j$ $s_2 = -2-4j$, $s_3 = -10$

step 1: rank $[B \mid AB \mid A^2B]$

$= \text{rank} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 31 \end{bmatrix} = 3$. 滿秩.

step 2: desired characteristic equation.

$(s-s_1)(s-s_2)(s-s_3)$

$= (s+2-4j)(s+2+4j)(s+10)$

$= s^3 + 14s^2 + 60s + 200$

step 3: $|sI-A+BK| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1+k_1 & 5+k_2 & s+6+k_3 \end{vmatrix} = s^3 + (6+k_3)s^2 + (5+k_2)s + 1+k_1$

$\Rightarrow \begin{cases} 6+k_3 = 14 \\ 5+k_2 = 60 \\ 1+k_1 = 200 \end{cases}$ $\Rightarrow K = [k_1 \ k_2 \ k_3] = [199, 55, 8]$

Control

$u = -K\vec{x} = -199x_1 - 55x_2 - 8x_3$.